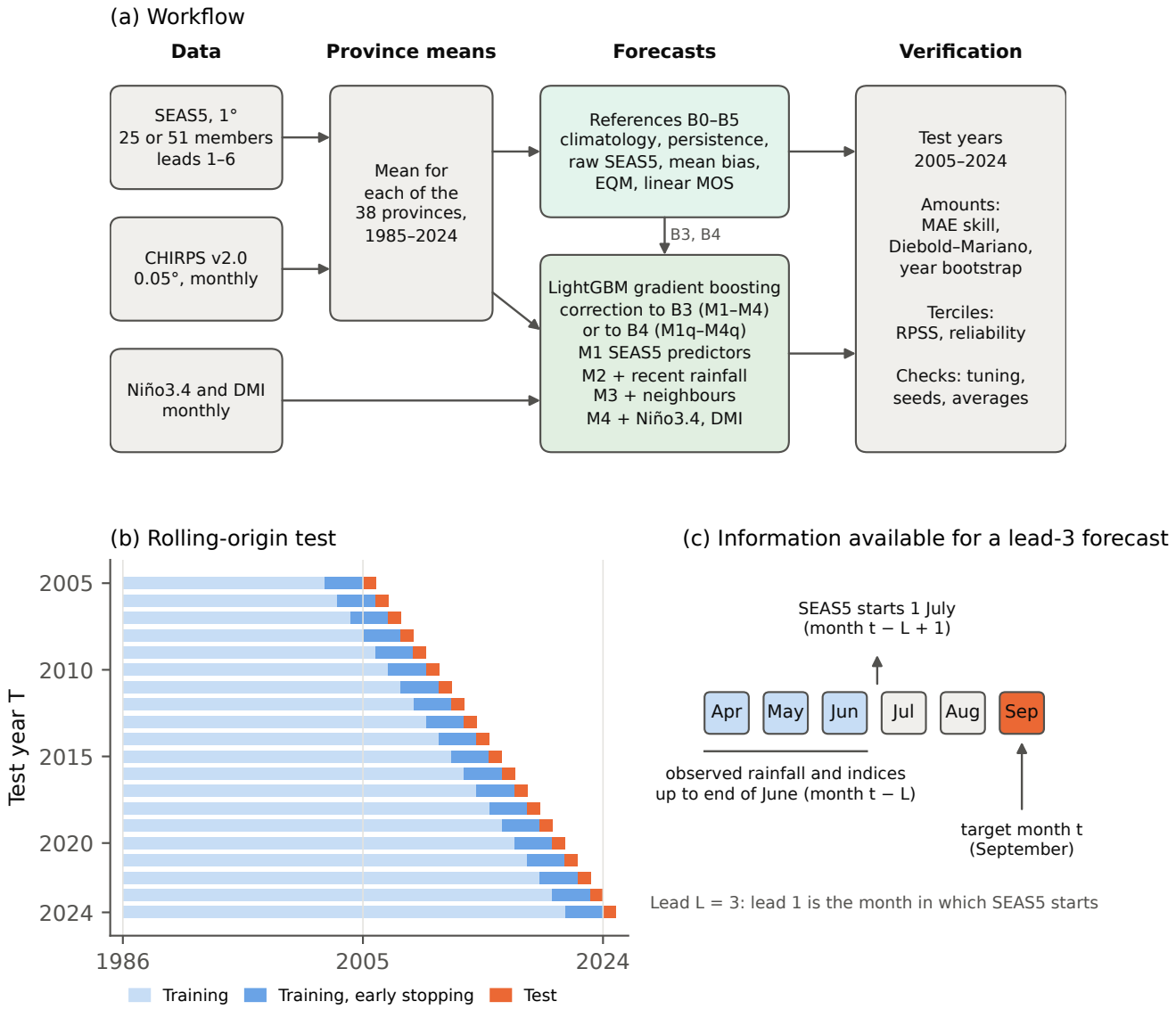
(a) Workflow
Data
Province means
Forecasts
Verification
SEAS5, 1°
25 or 51 members
leads 1–6
CHIRPS v2.0
0.05°, monthly
Niño3.4 and DMI
monthly
Mean for
each of the
38 provinces,
1985–2024
References B0–B5
climatology, persistence,
raw SEAS5, mean bias,
EQM, linear MOS
B3, B4
LightGBM gradient boosting
correction to B3 (M1–M4)
or to B4 (M1q–M4q)
M1 SEAS5 predictors
M2 + recent rainfall
M3 + neighbours
M4 + Niño3.4, DMI
Test years
2005–2024
Amounts:
MAE skill,
Diebold–Mariano,
year bootstrap
Terciles:
RPSS, reliability
Checks: tuning,
seeds, averages
(b) Rolling-origin test
Test year T
2005
2010
2015
2020
2024
1986
2005
2024
Training
Training, early stopping
Test
(c) Information available for a lead-3 forecast
SEAS5 starts 1 July
(month t − L + 1)
Apr
May
Jun
Jul
Aug
Sep
observed rainfall and indices
up to end of June (month t − L)
target month t
(September)
Lead L = 3: lead 1 is the month in which SEAS5 starts